

→ Machine Learning Statistics

→ Make sense of data, decision making using the data.

→ analyse data.

→ marketing, health care,

→ Descriptive: Organise the data you have,
not drawing any conclusions beyond the data.

mean, median, mode.

Variation, SD.

→ Inferential: Experiments, conclusions, predictions

Define data :-

Population

Sample

↓ Variable :- characteristic.

Experiment: conclusions drawn on a sample of data.

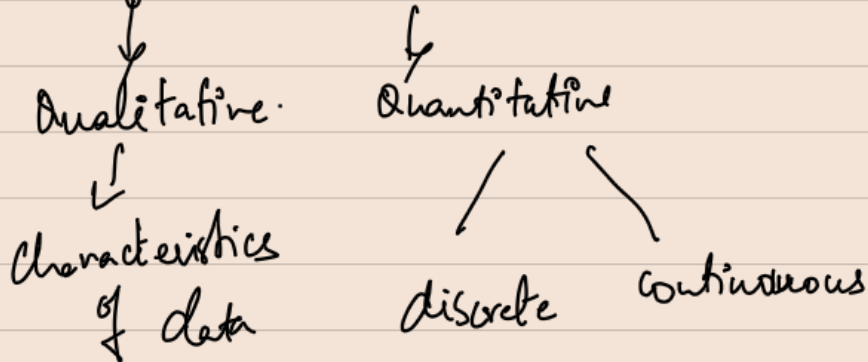
Parameters: defines / summarize data.

→ describe:- Info of data.

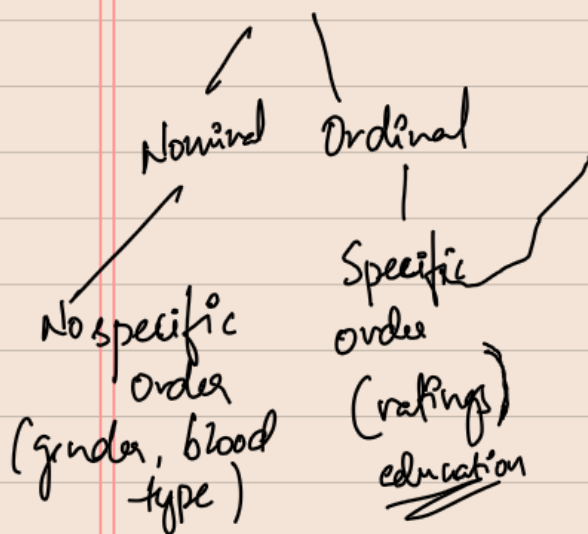
→ collection of facts, info

Types of data

→ Categorical, Numerical



Qualitative



Interval:- degree Celsius & Fahrenheit

Ratio:- degree Kelvin.

→ Measures of central tendency:-

mean is susceptible to outliers
so not a good measure of central tendency.

median :- Sorting the list and finding
the centre most value.

mode :- Mode is most occurring data.

→ Usually the good measure of central tendency.

→ Measure of dispersion.

↓
Variance, SD.

↓
How is the spread
around the second point



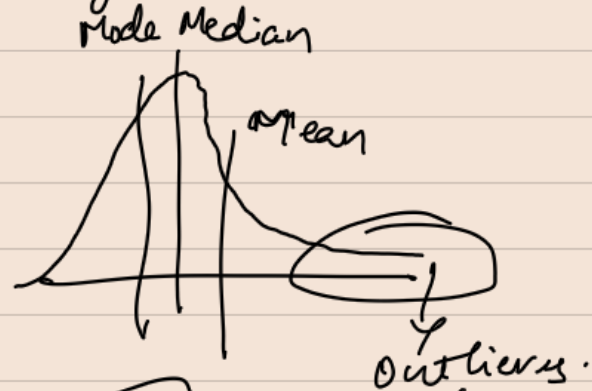
How each value far from variance.

$$S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} \rightarrow \text{makes the differences to positive.}$$

$$SD = \sqrt{\text{variance.}}$$

$$S = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$

Symmetrical and asymmetrical distribution.



left skewness

(or)
Right skewness

} → measures of symmetry.

Quantiles :-

$100/4$

25, 50, 75, 100

IQR : $q_3 - q_1$

lower bound : $q_1 - 1.5 * IQR$

upper bound : $q_3 + 1.5 * IQR$

} out of this then outlier.

$$Z\text{-score} = \frac{x - \text{mean}}{SD}$$